EXPERIMENT-24:

AIM:

To write a C program to determine the private key of an RSA system using the given public key \(e = 31\), \(n = 3599\), and the Extended Euclidean Algorithm.

**CODE:**

```
#include <stdio.h>


int main() {
   int n = 3599, e = 31, p, q, phi, d, i;


   for (i = 2; i * i <= n; i++) {
      if (n % i == 0) {
         p = i;
         q = n / i;
         break;
      }
   }


   phi = (p - 1) * (q - 1);


   for (d = 1; d < phi; d++)
      if ((e * d) % phi == 1)
         break;


   printf("p = %d\nq = %d\n", p, q);
   printf("Phi(n) = %d\n", phi);
   printf("Public key = (%d, %d)\n", e, n);
```

```
    printf("Private key = (%d, %d)\n", d, n);


    return 0;
}
```

**OUTPUT:**

